

Preferred Formats

For Dissemination and Preservation

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1. Preferred Formats - Guidance and Tools

The draft preferred formats guidance is based on content developed by DANS¹, which in turn is the result of a continuous monitoring and engagement effort by the DANS Data Curators.

Preferred formats are those of which one is confident that they will offer the best long-term guarantees in terms of usability, accessibility, and sustainability.

Format guidance has two separate objectives, and these do not always result in the same format recommendations:

1. For dissemination, current popular formats are often preferred, although these may not be optimally preservable.
2. For preservation, the main consideration is whether such formats will be human- and machine-readable in the long term in the absence of specific software.

In practice, it is not always possible to use formats which satisfy all of these criteria, but using them to the extent possible should always be considered, since this improves the interoperability and reusability of outputs.

¹



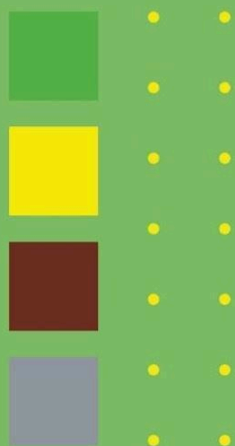
Table 1. Preferred Formats

Type	Long-Term Preservation	Dissemination	Recommendation
Documents	PDF/A (.pdf) ODT (.odt)	Microsoft Word (.doc) Office Open XML (.docx) Rich Text File (.rtf) PDF other than PDF/A (.pdf)	PDF/A (.pdf)
Plain text	Unicode text (.txt)	Non-Unicode text (.txt)	Unicode text (.txt)
Markup language	XML (.xml) HTML (.html) Related files: .css , .xslt , .js , .es	SGML (.sgml) Markdown (.md)	Any Unicode text file
Spreadsheets	ODS (.ods) CSV (.csv)	Microsoft Excel (.xls) Office Open XML Workbook (.xlsx) PDF/A (.pdf)	CSV (.csv) If CSV will result in loss of formula logic, use .xlsx
Databases	SQL (.sql) SIARD (.siard) CSV (.csv)	Microsoft Access (.mdb, .accdb) dBase (.dbf) HDF5 (.hdf5, .he5, .h5)	SQL (.sql) Export the SQL statements required to create and update all tables and views

Statistical data	SPSS (.dat/.sps) STATA (.dat/.DO) JASP (.csv/.html) R	SPSS Portable (.por) SPSS (.sav) STATA (.dta) SAS (.7dat; .sd2; .tpt) JASP (.jasp)	SPSS (.dat/.sps) STATA (.dat/.DO) JASP (.csv/.html) R
Raster images	JPEG (.jpg, .jpeg) TIFF (.tif, .tiff) PNG (.png) JPEG 2000 (.jp2) DICOM (.dcm)		JPEG (.jpg, .jpeg) TIFF (.tif, .tiff) PNG (.png) JPEG 2000 (.jp2) DICOM (.dcm)
Vector images	SVG (.svg)	Adobe Illustrator (.ai) EPS (.eps) WMF/EMF (.wmf, .emf) CDR (.cdr)	SVG (.svg)
Audio	BWF (.bwf) MXF (.mxf) Matroska (.mka) FLAC (.flac) OPUS	WAVE (.wav) MP3 (.mp3) AAC (.aac, .m4a) AIFF (.aif, .aiff) OGG (.ogg)	Matroska (.mka) Or MP3 (.mp3)

Video	MXF (.mxf) Matroska (.mkv)	MPEG-4 (.mp4, .m4a, .m4v, ...) MPEG-2 (.mpg, .mpeg, .m2v, .mpg2, ...) AVI (.avi) QuickTime (.mov, .qt)	Matroska (.mkv) Or MPEG-4 (.mp4)
3D	WaveFront Object (.obj) Polygon file format (.ply) X3D (.x3d) COLLADA (.dae)	Autodesk FBX (.fbx) Blender (.blend) 3D PDF (.pdf)	WaveFront Object (.obj)
RDF	RDF/XML (.rdf) Trig (.trig) Turtle (.ttl) NTriples (.nt) JSON-LD		RDF/XML (.rdf) Trig (.trig) Turtle (.ttl) NTriples (.nt) JSON-LD
Software	Place a copy of your git-based software repository in Software Heritage	Any git-based publicly accessible repository (GitHub, GitLab, SourceForge, Bitbucket).	GitHub (most commonly used) GitLab (non-commercial)

If you encounter any formats that are not listed, submit a ticket via the Helpdesk using the “Help” button at the bottom right of the kb-rda.org website.



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